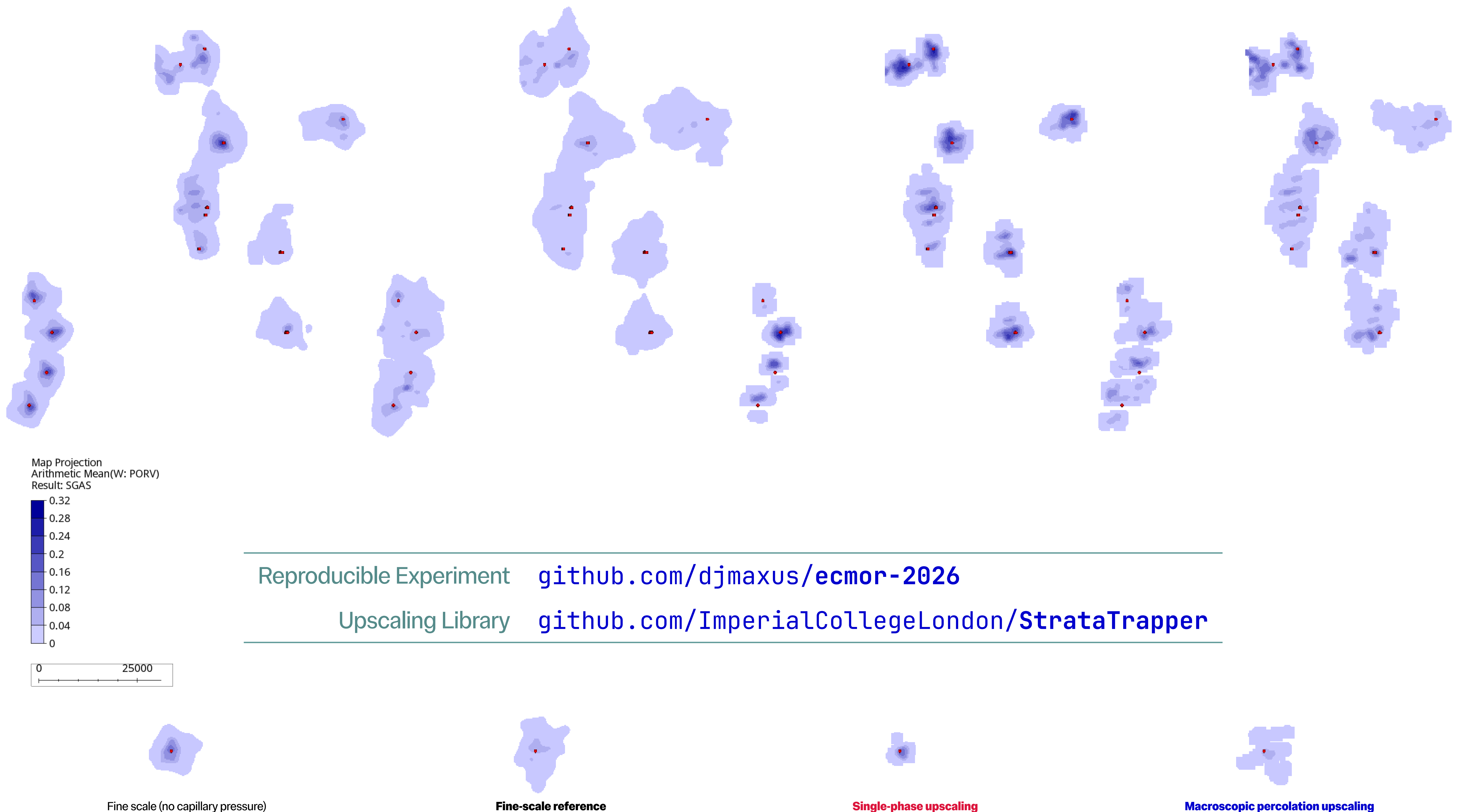


Capillary heterogeneity effects upscaled with macroscopic percolation for the Troll aquifer model

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Reproducible Experiment github.com/djmaxus/ecmor-2026

Upscaling Library github.com/ImperialCollegeLondon/StrataTrapper

Summary

Keywords: CCS · Geological CO₂ storage · dynamic reservoir simulation sub-grid capillary heterogeneity · accuracy-feasibility trade-off · capillary-limit upscaling Macroscopic Invasion Percolation · directional relative permeability · Troll aquifer model regional scale · vertical upscaling · parallel computing · OPM Flow · ResInsight · C++ codegen

Troll aquifer model

- Large regional model with faults
- 25 years of CO₂ injection, 14 wells
- 500 years of post-injection monitoring
- Simulated with **OPM Flow** (CO2STORE, ENDSALE, JFUNC, SGWFN, no imbibition)

Coarse mesh

- vertically upscaled from 217 to 5 layers
- 27 × fewer active cells

Capillary heterogeneity

Introduced to the model via Leverett-J

$$p_c(\bar{s}_w) = \sigma J(\bar{s}_w) \sqrt{\frac{2\varphi}{K_x + K_y}}$$

$J(\bar{s}_w)$ is weakened normalized capillary pressure from the Smeaheia dataset.

Macroscopic Invasion Percolation

- Static modelling tool
- Counterpart of pore-scale percolation
- Individual pores replaced with fine-scale cells
- Assumes local capillary equilibrium
- Simulates drainage of wetting phase
- Discrepancies in entry pressures yield steady-state fluid distributions

MIP Upscaling

Novel capillary-limit static upscaling. (see: Elizarev et al., 2025; Wolff et al., 2013)

Local, per-cell MIP

- Integrated MIP-controlled fluid content → **upscaled capillary pressure**
- Local Darcy flow w.r.t MIP → **upscaled directional phase permeabilities**

StrataTrapper ≥ v0.17.0

- OPM Flow input deck export
- automatic C++ code generation, 6 × upscaling acceleration

- (1) **Contribution to OPM Flow**
Support for directional scaling of relative permeabilities in gas-water models:
github.com/OPM/opm-common/pull/4923

Simulation Results

- Improved accuracy with MIP upscaling
- areal extent of CO₂ plumes, migration paths
- operational state of wells
- Preserved pressure match
- Performance gain: 2 orders of magnitude

Results

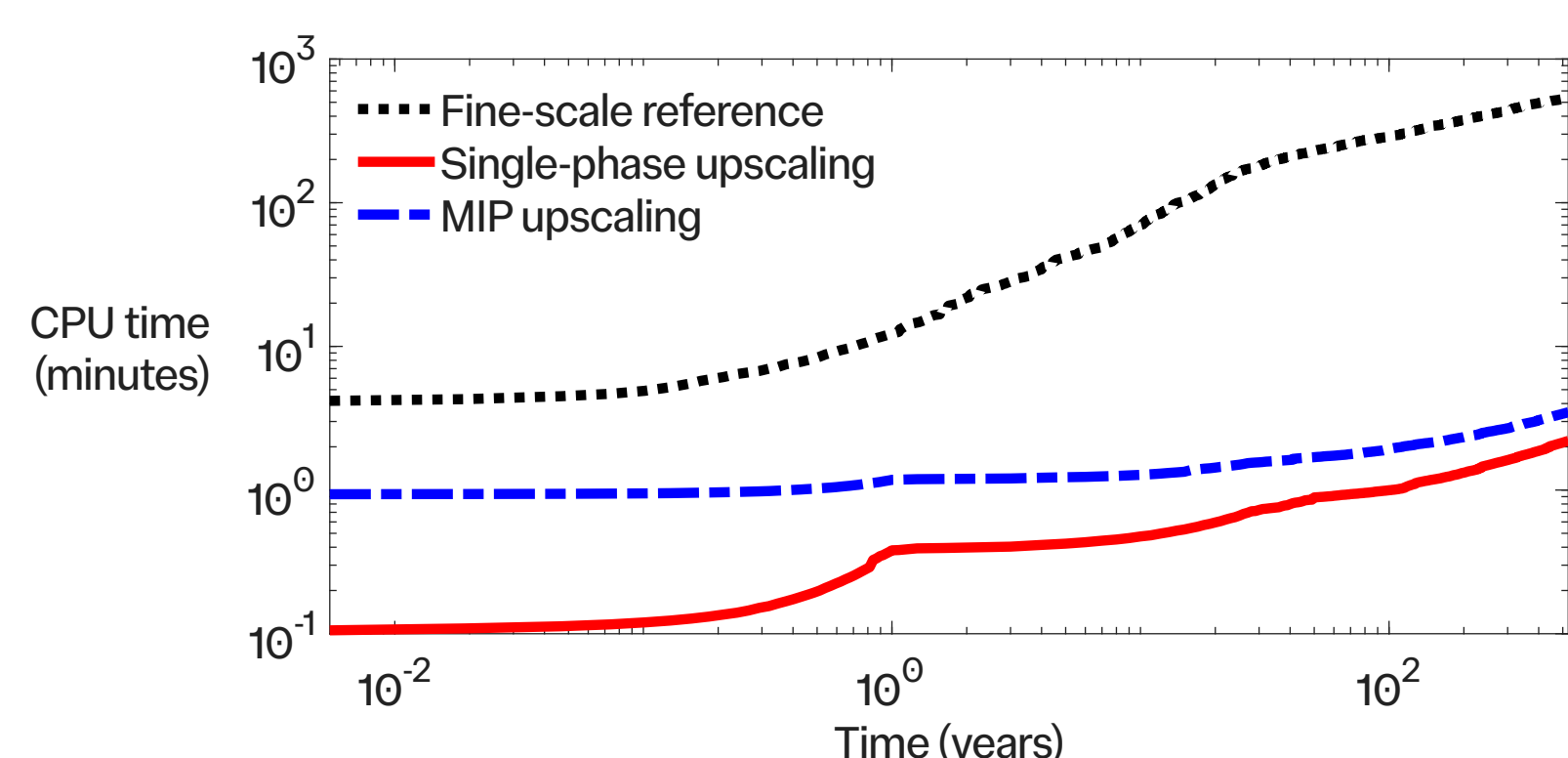


Figure 2: Computational performance of reservoir simulations with OPM Flow, 16 MPI nodes.

Figure 1: CO₂ saturations at the end of injection (ResInsight contour maps). Injection wells are indicated by red spheres.

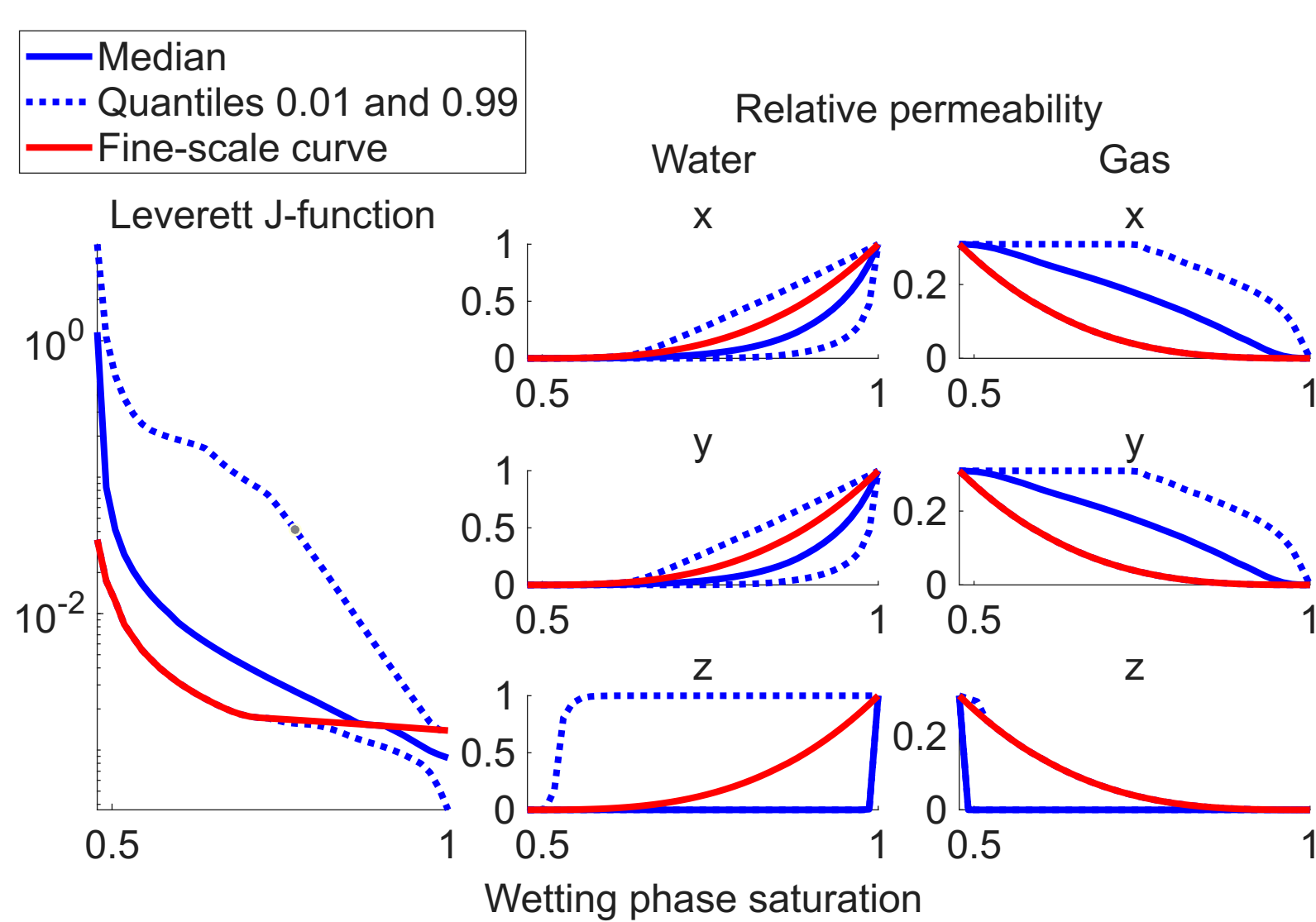


Figure 3: Variety of MIP-upscaled flow functions. Fine-scale inputs are in red. Statistical plots of MIP-upscaled functions are in blue.

Pressure match preserved

In the presence of capillary pressure $p_{c,gw}$, phase pressures are non-equivalent.

$$p_g = p_w + p_{c,gw}$$

Figure 8 of the extended abstract shows the pressure of gas p_g , which is the reference pressure in CO2STORE simulations. In the abstract, we did not provide a definitive explanation to the reservoir pressure discrepancy we observed.

Figure 4 of the mean reservoir water pressure p_w suggests that upscaled simulations stayed closely aligned in pressure, preserving the achievement of Landa-Marbán et al. (2025).

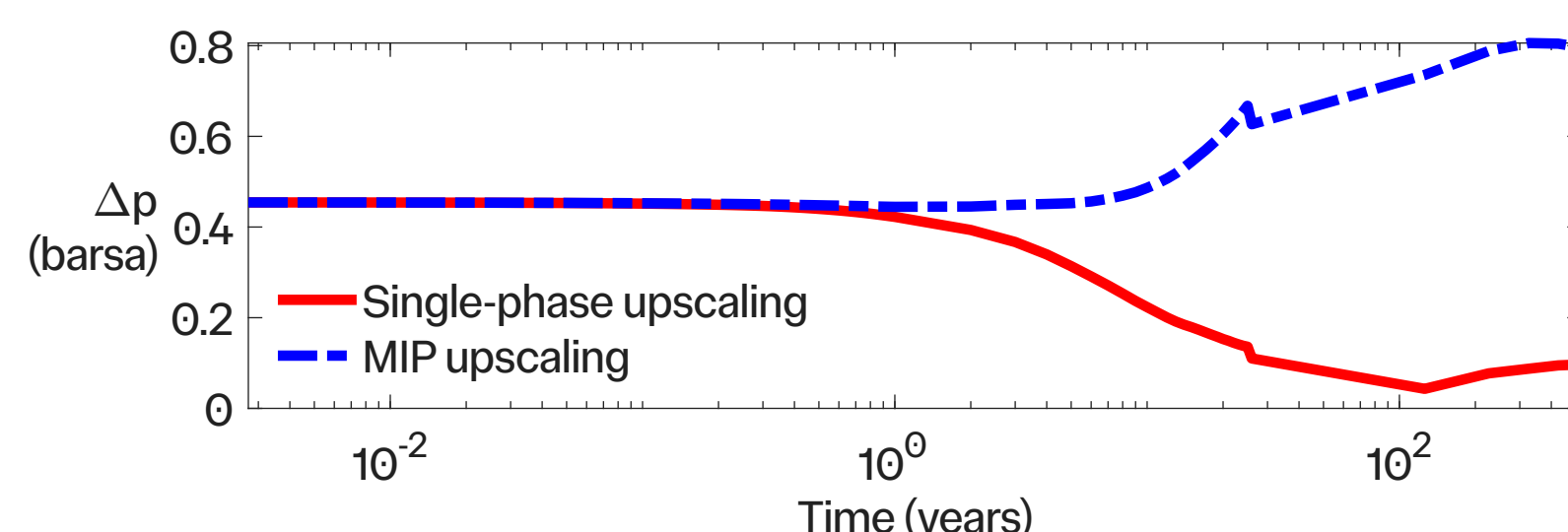


Figure 4: Water phase pressure dynamics in upscaled simulations. Log-scale time, volume-weighted average, difference with reference fine-scale simulation.

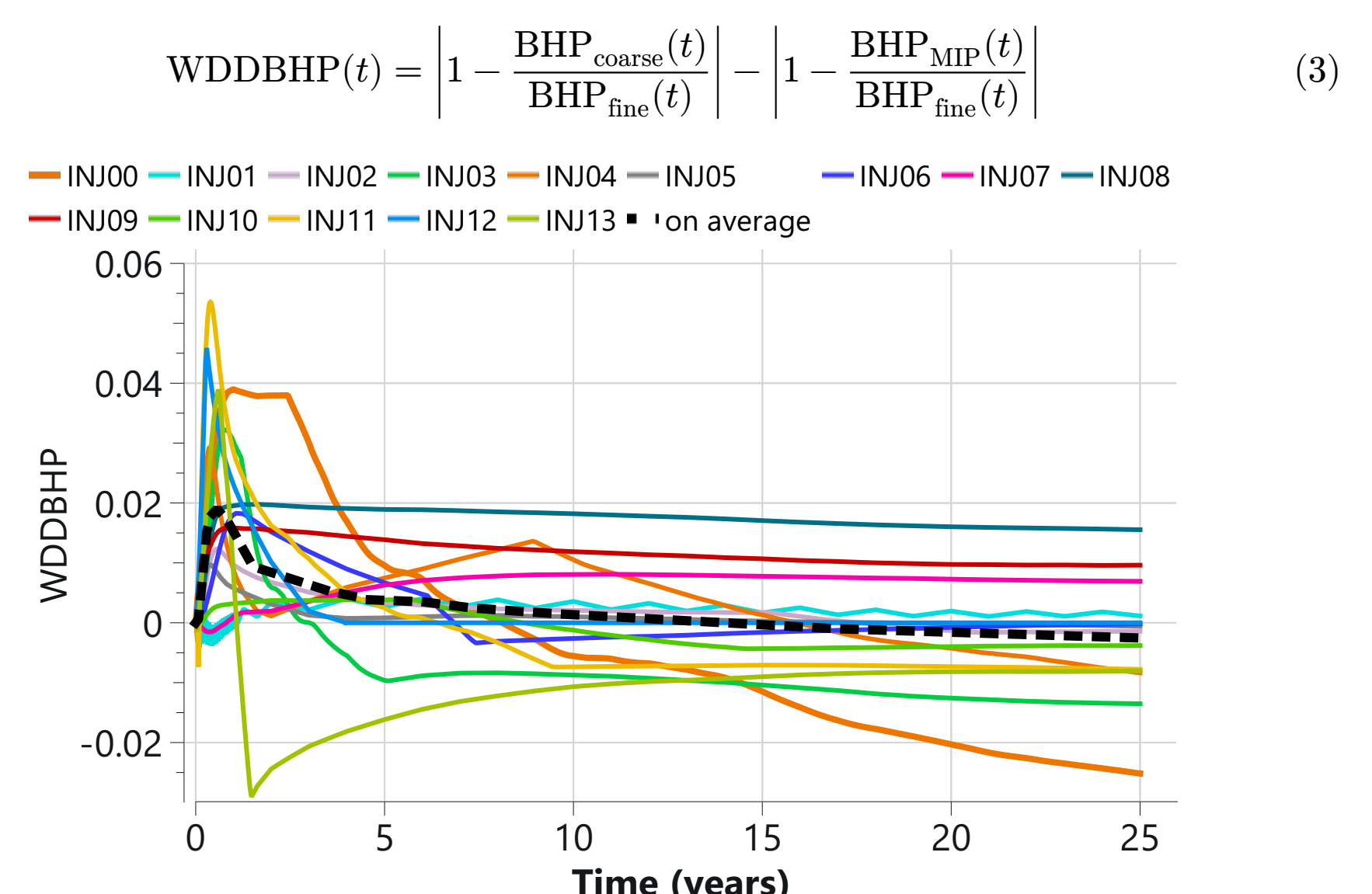
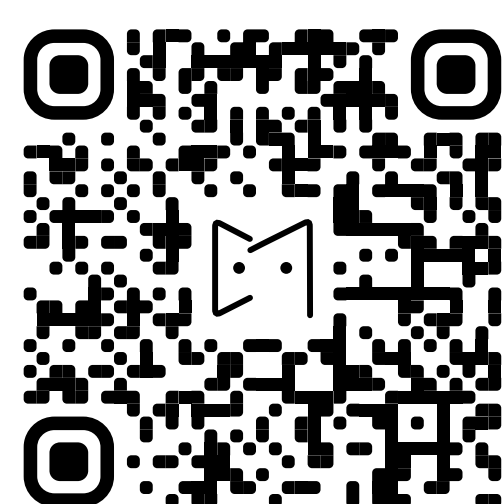


Figure 5: Relative improvement in bottom-hole pressure prediction for the injection wells as per Equation 3

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An adaptive multiscale approach for modeling two-phase flow in porous media including capillary pressure

Acknowledgements

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